Endogenous Education Choice in Segmented Search Markets

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Abstract

How do workers decide whether to acquire education? The standard answer weighs wages against costs, but this misses a deeper aspect of the choice: education is not just about whether the skilled market pays more, it is about whether that market will actually have jobs available after training that were worth the trouble. This paper develops a general equilibrium search model in which the fraction of workers who choose to train is determined endogenously alongside wages, unemployment, and market tightness in both markets. Workers differ in ability, which governs the cost of training; the skilled market additionally features on-the-job search and a job ladder, giving trained workers richer career dynamics than their unskilled counterparts. The model is estimated on U.S. data across multiple aggregate states of the economy using Simulated Method of Moments. This allows me to trace how the economy — and the education margin — adjusts over time after an aggregate shock. I follow this with two applications that demonstrate the model's usefulness: the first studies the effect of more generous outside options (UBI or unemployment insurance) on unemployment duration and match quality; the second compares two education subsidy designs: flow transfers during training versus upfront cost reductions.

1 Data and Estimation

1.1 Data

The model is estimated on U.S. data from four sources. Labour-market stocks and worker transition rates are constructed from the Current Population Survey—Basic Monthly Files (CPS). Wage moments are drawn from the CPS Annual Social and Economic Supplement (ASEC). The skilled employer-to-employer transition rate comes from the Census Bureau's Job-to-Job Flows (J2J) data, which provides quarterly counts of hires from other employers by education group. Vacancy data for market tightness come from the Job Openings and Labor Turnover Survey (JOLTS).

The sample consists of people (civilians, military was excluded) aged 25–54 who report positive labour-force attachment. I partition workers into two skill groups: those without a bachelor's degree (unskilled) and those with a bachelor's degree or above (skilled). This partition is the empirical counterpart of the model's two-market structure. Monthly transition rates—unemployment to employment (UE), employment to unemployment (EU), and employer to employer (EE)—are constructed from matched CPS records using standard methods.

Moments are computed over four windows, each chosen to approximate a stationary regime:

Table 1: Estimation windows

Period	Dates	Length
Baseline fc	Jan 2003 – Dec 2007	60 months
Crisis fc	Jan 2008 – Jun 2009	18 months
Baseline covid	Jan 2015 – Dec 2019	60 months
Crisis covid	Mar 2020 – Dec 2021	22 months

The two baseline windows cover the expansion phases preceding each crisis. The crisis windows cover the contraction through the trough. Within each window, I average the monthly observations to obtain a single vector of target moments and treat this average as the stationary equilibrium that the model should match.

1.2 Moments and Identification

I estimate the model by the Simulated Method of Moments in a standard fashion. The estimation targets 23 moments grouped into four blocks. Table 2 lists each moment with its model counterpart.

Table 2: Targeted moments

Moment	Model counterpart	Source
Labour-market stocks		
Skilled share	$\int_{\bar{x}}^{1} [u_S(x) + \int e_S(x,p)\, dp]\, dx$	CPS
Training share	$\int_{\bar{x}}^{1} t(x)\, dx$	CPS
Variance of unskilled wages	$\mathrm{Var}[w_U]$	CPS ASEC
Skewness of unskilled wages	$\mathrm{CM}_3[w_U]$	CPS ASEC
Variance of skilled wages	$\mathrm{Var}[w_S]$	CPS ASEC
Skewness of skilled wages	$\mathrm{CM}_3[w_S]$	CPS ASEC
Transition rates		
Job-finding rate, unskilled	$\kappa_U(z)\,[1 - G_z(p_U^*)]$	CPS
Separation rate, unskilled	$\lambda_U(z)\, G_z(p_U^*)$	CPS
Job-finding rate, skilled	$\kappa_S(z)\,[1 - \Gamma_z(p_S^*)]$	CPS
Separation rate, skilled	$\xi_S(z) + \lambda_S(z)\,\Gamma_z(p_S^*)$	CPS
EE rate, skilled	$\kappa_S(z)\,[1 - \Gamma_z(p_S^{oj})]$	J2J
Wages		
Mean wage, unskilled	$\mathbb{E}[w_U(x,p,z)]$	CPS ASEC
Mean wage, skilled	$\mathbb{E}[w_S(x,p,z)]$	CPS ASEC
25th percentile wage, unskilled	$Q_{0.25}[w_U]$	CPS ASEC
25th percentile wage, skilled	$Q_{0.25}[w_S]$	CPS ASEC
Median wage, unskilled	$Q_{0.50}[w_U]$	CPS ASEC
Median wage, skilled	$Q_{0.50}[w_S]$	CPS ASEC
Log skill premium	$\mathbb{E}[\ln w_S] - \mathbb{E}[\ln w_U]$	CPS ASEC
Tightness		
θ_U	V_U/U_U	JOLTS / CPS
θ_S	V_S/U_S^{eff}	JOLTS / CPS

The choice of moments reflects the structure of the model. The skilled share and training share directly discipline the training cutoff $\bar{x}(z)$. The cross-sectional wage dispersion moments—variance and skewness of the wage distribution within each skill group—pin down the shape parameters of the match-quality distributions and help separate the ability distribution from the match-quality distribution. Without these moments, the two sources of wage variation would be poorly identified separately.

Transition rates play the central role in identifying market-specific frictions. The unskilled UE rate identifies the contact rate $\kappa_U(z)$, and therefore—through the matching function and free-entry condition—the matching efficiency and vacancy cost. The unskilled EU rate identifies the shock arrival rate $\lambda_U(z)$ and the reservation quality $p_U^*(x, z)$: given the production function, these two objects are tightly linked. On the skilled side, the same logic applies, with the additional information from the employer-to-employer (EE) transition rate. This moment is particularly valuable because it identifies the intensity of on-the-job search: it pins down the contact rate $\kappa_S(z)$ jointly with the on-the-job search cutoff $p_S^{oj}(x, z)$ and helps separate the offer distribution Γ_z from the separation threshold $p_S^*(x, z)$.

Wage levels identify the productivity shifters $P_U(z)$ and $P_S(z)$. The log skill premium identifies the return to training. Wage quantiles—particularly the gap between the 25th percentile and the median—are informative about the shape of the match-quality distributions and the bargaining shares β_j: a compressed lower tail relative to the median suggests a tight match-quality distribution or high bargaining power, while a long lower tail suggests the opposite.

Market tightness $\theta_j(z)$ directly constrains vacancy costs $k_j(z)$ through the free-entry conditions. Without these moments, vacancy costs would be identified only indirectly through their effect on contact rates, which also depend on matching efficiencies.

1.3 Estimation

The SMM estimator minimises the weighted distance between the data moments $\hat{m}$ and their model counterparts $m(\theta)$:

$$\hat{\theta} = \arg\min_{\theta} \big[\hat{m} - m(\theta)\big]' \, W \, \big[\hat{m} - m(\theta)\big], \tag{1}$$

where W is the optimal weight matrix, constructed as the inverse of the estimated variance–covariance matrix of $\hat{m}$. I estimate $\hat{\Sigma}$ using influence-function standard errors with shrinkage applied to off-diagonal elements when the matrix is ill-conditioned; details are in Appendix **??**.

The model moments $m(\theta)$ are computed by solving for the stationary equilibrium at parameter vector θ and evaluating the equilibrium objects described in Section **??**.

Parameter classification

Three parameters—the discount rate r, the demographic exit rate ν, and the training completion rate ϕ—are calibrated externally and held fixed across all estimation windows. Table 3 reports their values.

Table 3: Externally calibrated parameters

Parameter	Source	Value
r	Monthly discount rate ($\approx 5\%$ annual)	0.0042
ν	CPS matched-panel labour-force exit hazard	0.0384
ϕ	National Student Clearinghouse completion rate	0.0222

A second group of parameters is fixed externally to simplify the model. Firms and workers share surplus equally and the matching function depends symmetrically on unemployment and vacancies, so the bargaining shares (β_U, β_S) and matching elasticities (η_U, η_S) are all set to 0.5. The OJS flow cost is likewise fixed, at $\sigma_S = 1.1$.

The remaining parameters are estimated, and fall into two groups. *Deep structural parameters* reflect preferences, institutions, and technology that do not change with the aggregate state: the ability distribution (a_ℓ, b_ℓ), the training cost c, the flow payoffs (b_U, b_T, b_S), and the matching efficiencies (μ_U, μ_S). *Regime-specific parameters* capture features of the aggregate state that shift between regimes: the sector productivities (P_U, P_S), the match-quality distribution parameters $(\alpha_U, a_\Gamma, b_\Gamma)$, the vacancy costs (k_U, k_S), the shock arrival rates (λ_U, λ_S), and the exogenous skilled separation hazard ξ_S.

Multi-regime estimation strategy

The estimation proceeds in two stages. In the first stage, I estimate the model on the baseline window, treating all deep structural and regime-specific parameters as free. This stage identifies the full parameter vector from a single cross-section of moments.

In the second stage, I fix the deep structural parameters at their baseline estimates and re-estimate only the regime-specific parameters on the crisis window. This procedure identifies what changed during the crisis: which parameters shifted, by how much, and in which direction. The identifying assumption is that preferences, matching technology, and bargaining institutions did not change between the baseline and the crisis—only the aggregate state did.

The same two-stage procedure is applied separately to the financial-crisis moments pair (baseline fc → crisis fc) and the COVID pair (baseline covid → crisis covid). Comparing the deep structural estimates across the two baselines provides a structural stability check: if the deep parameters are similar across the two expansions, the identifying assumption that they are regime-invariant is internally consistent.

2 Results

This section presents the estimation results for the financial-crisis episode. I begin with the baseline equilibrium estimated on the pre-crisis expansion (2003–2007), examine the model's internal structure by simulating it for the estimated parameters, assess the quality of the moment fit, and then turn to the crisis re-estimation (2008–2009) to identify which features of the aggregate state shifted during the downturn.

2.1 Baseline Equilibria

The model is estimated on the baseline financial crisis window and solved at the estimated parameters. I now describe the resulting equilibrium objects.

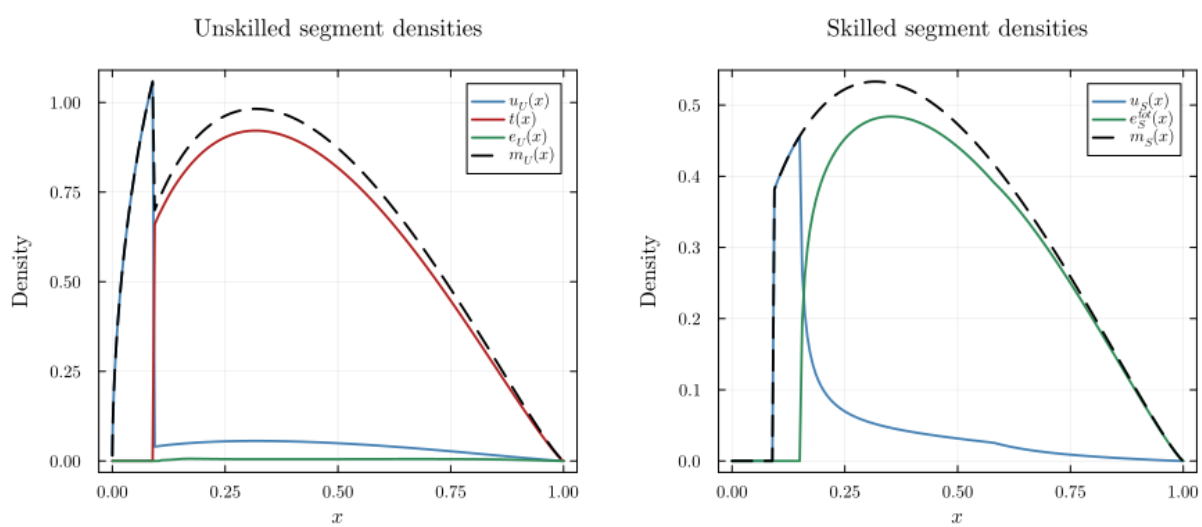

Figure 1: Cross-sectional densities over worker type x: unskilled segment (left) and skilled segment (right). The training cutoff $\bar{x}$ partitions the two markets.

Cross-sectional distributions. The figure shows the endogenous segmentation of the economy. The training cutoff partitions the ability distribution into two pools with different mean characteristics. Above $\bar{x}$, the mass flows through the training pipeline into skilled

unemployment and then into skilled employment. Below $\bar{x}$, workers remain in the unskilled market permanently.

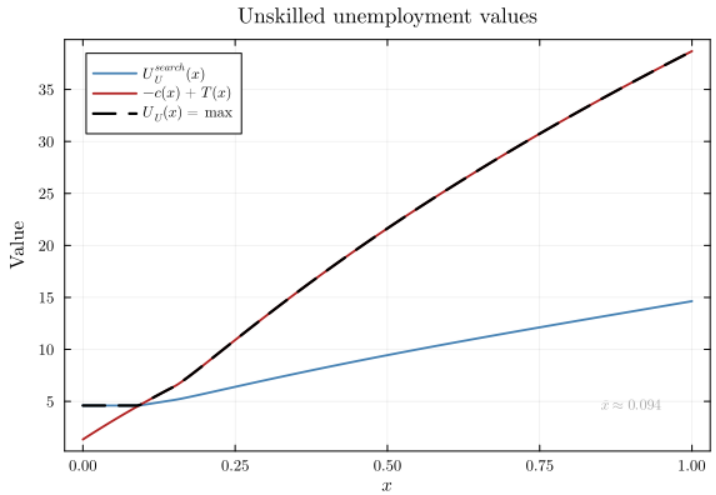

Figure 2: Unskilled unemployment values

Training decision. For low-ability workers, the value of unskilled search exceeds the net value of training: the training cost is too high and the expected gains from the skilled market too low. For high-ability workers, the net value of training dominates: training costs are low, and the continuation value of the skilled market—including job-ladder gains from on-the-job search—is high. The gap between the two curves at $\bar{x}$ is zero by construction; above $\bar{x}$ it widens, reflecting the increasing net gain from training for higher-ability types.

Surplus and separation rules. The surplus maps summarise the equilibrium in each market. Sorting is present in both markets. A key comparative static is the slope of the reservation cutoff in x: flatter curves mean that the training threshold has a weaker effect on match stability.

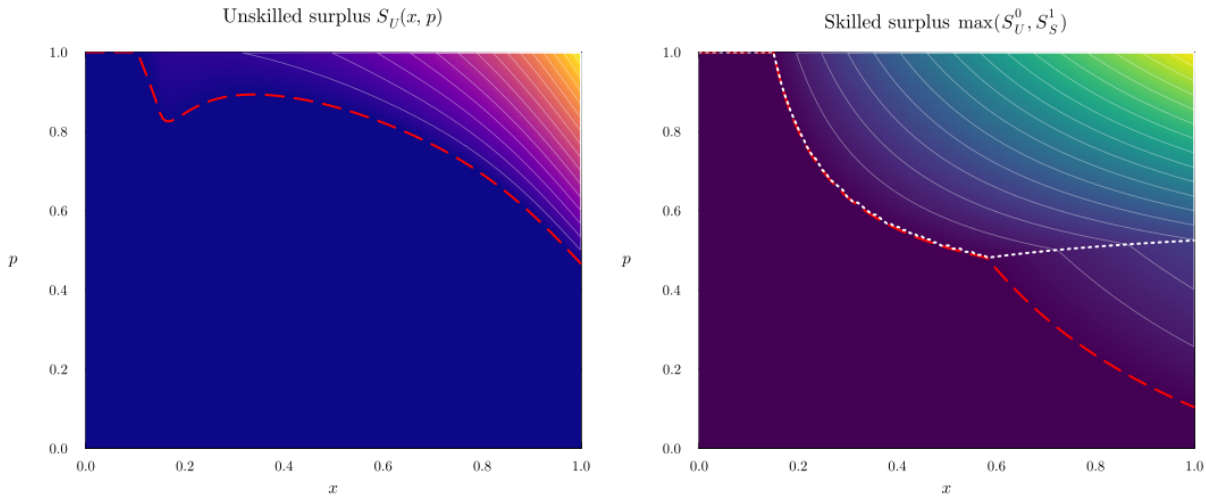

Figure 3: Match surplus over the (x, p) plane: unskilled surplus $S_U(x, p)$ (left) and skilled surplus $\max(S_S^0, S_S^1)$ (right). Overlaid curves mark the reservation quality $p_j^*(x)$ and, in the skilled panel, the on-the-job search cutoff $p_S^{oj}(x)$.

2.2 Moment Fit

Tables 4 and 5 report the model fit for both baseline windows—the pre-FC expansion (2003–2007) and the pre-COVID expansion (2015–2019)—placing data moments alongside their model counterparts and the difference between them. The two baselines are shown side by side to facilitate the structural stability comparison: the deep parameters are estimated on the FC baseline and held fixed when the model is re-solved on the COVID baseline, so the COVID column reveals how well the same preference, technology, and institutional parameters rationalise a second, independent expansion twelve years later.

Wages. The wage block is the best-matched component of the estimation. For the FC baseline, mean wages, medians, and 25th-percentile wages are all within 1–9 percentage points of their data targets, and several—particularly the median and 25th-percentile of skilled wages—are matched almost exactly (differences of -0.003 and -0.002, respectively). The skill premium is slightly over-predicted in the FC window ($+0.09$) but nearly exact in the COVID window ($+0.01$). The COVID baseline shows somewhat larger deviations in skilled wage levels (mean wage skilled: -0.14; median skilled: -0.11), suggesting that the deep parameters estimated on the FC expansion generate a slightly compressed skilled wage distribution when applied to the tighter 2015–2019 labour market. The unskilled wage moments degrade less across baselines, consistent with the simpler structure of the unskilled block, where matches start at frontier quality and only deteriorate.

Table 4: Model Fit (A): Labour-Market Stocks and Transition Rates

Moment	Financial Crisis			COVID		
	Data	Model	Diff.	Data	Model	Diff.
Labour-market stocks						
exp(UR total)	1.0543	1.2274	+0.1731	1.0462	1.3041	+0.2579
	(0.0057)	(0.0025)		(0.0066)	(0.0019)	
exp(UR unskilled)	1.0661	1.2321	+0.1661	1.0582	1.3648	+0.3066
	(0.0063)	(0.0025)		(0.0086)	(0.0026)	
exp(UR skilled)	1.0264	1.2171	+0.1907	1.0250	1.1496	+0.1246
	(0.0043)	(0.0026)		(0.0027)	(0.0013)	
Skilled share	0.2915	0.3133	+0.0218	0.3596	0.2652	-0.0945
	(0.0208)	(0.0011)		(0.0295)	(0.0007)	
Training share	0.0334	0.5416	+0.5081	0.0450	0.4584	+0.4134
	(0.1061)	(0.0018)		(0.0903)	(0.0011)	
Var(w) unskilled	0.2543	0.1578	-0.0965	0.2860	0.1555	-0.1305
	(0.0643)	(0.0259)		(0.0432)	(0.0371)	
CM3(w) unskilled	0.1709	0.0110	-0.1599	0.2624	0.0402	-0.2222
	(0.0740)	(0.0124)		(0.0454)	(0.0394)	
Var(w) skilled	0.3934	0.2611	-0.1323	0.4498	0.3023	-0.1475
	(0.0477)	(0.0315)		(0.0480)	(0.0402)	
CM3(w) skilled	0.2337	0.0980	-0.1357	0.3514	0.1400	-0.2114
	(0.0470)	(0.0359)		(0.0320)	(0.0272)	
Transition rates						
JFR unskilled	0.2610	0.0083	-0.2528	0.2565	0.0639	-0.1926
	(0.0849)	(0.0177)		(0.0846)	(0.0099)	
Sep. rate unskilled	0.0151	0.1097	+0.0945	0.0132	0.0241	+0.0109
	(0.0634)	(0.0384)		(0.1165)	(0.0155)	
JFR skilled	0.2741	0.3907	+0.1166	0.2717	0.4577	+0.1860
	(0.1500)	(0.0704)		(0.1398)	(0.0956)	
Sep. rate skilled	0.0062	0.0628	+0.0566	0.0059	0.0428	+0.0369
	(0.1246)	(0.0727)		(0.1617)	(0.0326)	
EE rate skilled	0.0080	0.0052	-0.0027	0.0085	0.0032	-0.0053
	(0.0624)	(0.0624)		(0.0366)	(0.0366)	

Table 5: Model Fit (B): Wages and Tightness

Moment	Financial Crisis			COVID		
	Data	Model	Diff.	Data	Model	Diff.
Wages						
Mean wage unskilled	0.9265	0.8493	-0.0772	0.9166	0.8030	-0.1136
	(0.0360)	(0.0178)		(0.0315)	(0.0269)	
Mean wage skilled	1.3239	1.2951	-0.0289	1.3144	1.1705	-0.1440
	(0.0346)	(0.0208)		(0.0289)	(0.0258)	
p25 wage unskilled	0.5652	0.4709	-0.0943	0.5482	0.4779	-0.0703
	(0.0393)	(0.0190)		(0.0453)	(0.0392)	
p25 wage skilled	0.8800	0.8776	-0.0024	0.8411	0.7201	-0.1210
	(0.0392)	(0.0231)		(0.0278)	(0.0264)	
p50 wage unskilled	0.8244	0.8371	+0.0127	0.7920	0.7337	-0.0583
	(0.0384)	(0.0194)		(0.0338)	(0.0285)	
p50 wage skilled	1.2042	1.2008	-0.0033	1.1594	1.0480	-0.1114
	(0.0377)	(0.0237)		(0.0298)	(0.0257)	
Skill premium	0.3846	0.4743	+0.0897	0.3851	0.3959	+0.0108
	(0.0136)	(0.0113)		(0.0172)	(0.0141)	
Tightness						
θ_U	0.9244	0.0367	-0.8878	1.5849	5.6596	+4.0747
	(0.2101)	(0.0880)		(0.2524)	(0.0567)	
θ_S	1.7020	0.9972	-0.7048	2.8943	1.3089	-1.5854
	(0.2508)	(0.1374)		(0.1508)	(0.0555)	

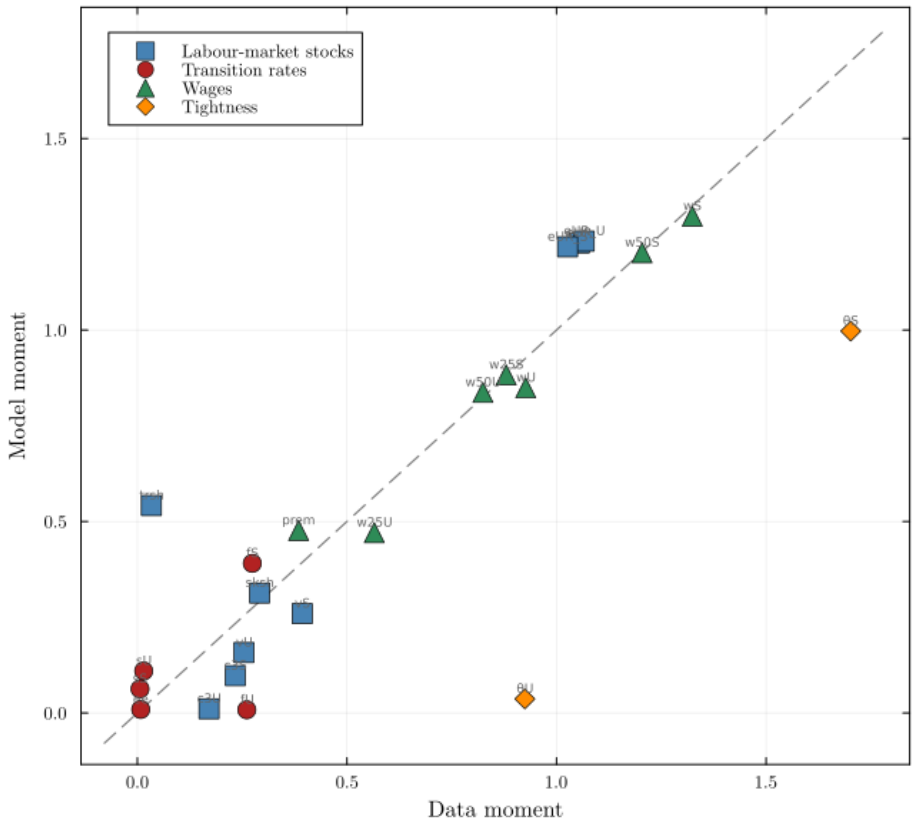

Figure 4: Model fit scatter

Labour-market stocks. Unemployment rates in exponentiated form are over-predicted in both baselines, with the discrepancy larger for the COVID window (total: +0.26 vs. +0.17; unskilled: +0.31 vs. +0.17). The skilled unemployment rate, however, is closer to the data in the COVID baseline (+0.12) than in the FC baseline (+0.19), indicating that the model captures the differential tightening of the skilled market during the later expansion more effectively than it captures the corresponding unskilled dynamics.

The skilled share is well matched in the FC baseline (+0.02) but under-predicted for the COVID baseline (−0.09). This is consistent with the secular rise in educational attainment between the two periods—a trend the model cannot generate, since the deep parameters governing the type distribution (a, b) and training cost c are held fixed at FC baseline values. A time-varying composition shift in the population, operating outside the model's regime-specific channels, likely accounts for the remainder.

The training share is the largest point of disagreement in both baselines: the model generates

a share of approximately 0.46–0.54, while the data—based on CPS college-enrolment flags for individuals without a BA—shows only 0.03–0.05. The data standard error on this moment is itself very large (0.09–0.11), reflecting the difficulty of measuring training stocks from cross-sectional survey indicators. Three factors contribute to the discrepancy. First, the model's training state captures all workers who have initiated the pipeline toward the skilled market, regardless of whether they would be classified as "enrolled in college" by the CPS. Second, the CPS measure excludes part-time and non-degree enrolments that may still represent human-capital investment in the model's sense. Third, with only a single training-cost parameter c disciplining this moment, the estimator trades off training-share fit against the other 22 moments in the objective, and the large data variance reduces the weight the optimal matrix places on this moment. The over-prediction should be borne in mind when interpreting the model's training cutoff $\bar{x}(z)$ and the implied mass flowing through the training pipeline.

Wage dispersion. The model systematically under-predicts the variance and third central moment of wages in both skill groups and both baselines. The under-prediction is more pronounced for the third central moment (skewness), particularly in the COVID baseline (CM3 skilled: -0.21; CM3 unskilled: -0.22). In the model, within-group wage dispersion arises from two sources: heterogeneity in innate type x and dispersion in match quality p. The compressed model variance suggests that the estimated match-quality distributions are tighter than what the data warrant, or that the model lacks an additional source of residual wage dispersion—such as within-match human capital accumulation or firm heterogeneity—that would fatten the tails of the cross-sectional wage distribution. The gap between data and model skewness, in particular, points toward a missing right tail: the model cannot generate the long upper tail of the wage distribution without allowing for firm-side heterogeneity or on-the-job skill growth, both of which are abstracted away.

Transition rates. The transition-rate block presents the most challenging fit. The unskilled job-finding rate (JFR) is severely under-predicted in both baselines (FC: 0.008 vs. 0.261; COVID: 0.064 vs. 0.257), indicating that the model's equilibrium unskilled contact rate—governed by matching efficiency μ_U, tightness θ_U, and the matching elasticity—is too low. Conversely, the unskilled separation rate is over-predicted in the FC baseline (+0.09) but nearly matched during the COVID baseline (+0.01), suggesting that the estimated shock-arrival rate λ_U and reservation rule $\underline{p}_U(x, z)$ are reasonable for the later expansion but generate excessive turnover when applied to the earlier one.

In the skilled market, the pattern reverses: the JFR is over-predicted (+0.12 FC; +0.19 COVID), while the separation rate is over-predicted by a smaller margin (+0.06 FC; +0.04

COVID). The skilled EE rate—the moment that identifies on-the-job search intensity—is well matched in both windows (differences of -0.003 and -0.005), confirming that the job-ladder mechanism and the OJS cutoff $p_S^{oj}(x, z)$ are appropriately disciplined. The opposing signs of the JFR discrepancies across the two markets reflect a tension inherent in the model: with matching efficiencies μ_U, μ_S held fixed across regimes, the estimator cannot simultaneously generate high contact rates in one market and moderate contact rates in the other without adjusting tightness, which is itself constrained by the free-entry conditions.

Market tightness. Tightness is the least well matched set of moments. In the FC baseline, both θ_U and θ_S are under-predicted (-0.89 and -0.70, respectively), whereas in the COVID baseline θ_U is dramatically over-predicted ($+4.07$) while θ_S remains under-predicted (-1.59). The sign reversal for unskilled tightness across the two baselines is notable: the FC deep parameters imply vacancy costs and matching efficiency levels that, when confronted with the much tighter 2015–2019 labour market, generate an equilibrium with far too many vacancies relative to the unskilled unemployed pool. This is consistent with the secular decline in the matching function's efficiency documented in the macro-labour literature, a time trend that the model—which treats μ_U as a fixed deep parameter—cannot accommodate. The discrepancy in skilled tightness is smaller in relative terms and consistent in sign across baselines, suggesting that the skilled free-entry condition is better anchored.

Cross-baseline assessment. Taken together, the moment fit delivers a clear hierarchy: wages are matched tightly, stock moments are matched at an intermediate level, and transition rates and tightness are matched least precisely. This ordering is consistent with the structure of the estimation—wage moments are pinned down by productivity shifters P_U, P_S with relatively few competing parameters, while transition rates depend on the interaction of tightness, matching efficiency, shock arrival, and endogenous separation, creating more scope for trade-offs in the objective function. The COVID baseline shows somewhat larger deviations than the FC baseline on average, which is expected: the deep parameters were estimated to fit the FC expansion and are only held fixed—not re-optimised—for the COVID window. That the wage fit remains strong and the skilled EE rate remains accurate across both baselines provides the core evidence for the structural stability of the deep parameters, supporting the identifying assumption that preferences and matching technology did not change between the two expansions.

Table 6: Parameter Estimates (A): Calibrated and Deep Structural

		Financial Crisis			COVID		
Parameter	Description	Baseline	Crisis	Δ	Baseline	Crisis	Δ
Externally calibrated							
r	Discount rate	0.00417	0.00417	—	0.00417	0.00417	—
ν	Demographic exit rate	0.03841	0.03841	—	0.03841	0.03841	—
φ	Training completion rate	0.02222	0.02222	—	0.02222	0.02222	—
Deep structural							
a_ℓ	Worker type shape	1.50538	1.50538	—	1.06876	1.50538	—
b_ℓ	Worker type shape	2.87219	2.87219	—	2.56120	2.87219	—
c	Training cost coeff.	2.94357	2.94357	—	2.91536	2.94357	—
b_U	Unskilled UI flow	0.19600	0.19600	—	0.23613	0.19600	—
b_T	Training flow	1.02176	1.02176	—	1.11815	1.02176	—
b_S	Skilled UI flow	0.56562	0.56562	—	0.43460	0.56562	—
μ_U	Unskilled matching eff.	0.04330	0.04330	—	0.02687	0.04330	—
η_U	Unskilled matching elas.	0.50000	0.50000	—	0.50000	0.50000	—
β_U	Unskilled bargaining	0.50000	0.50000	—	0.50000	0.50000	—
μ_S	Skilled matching eff.	1.18859	1.18859	—	0.84933	1.18859	—
η_S	Skilled matching elas.	0.50000	0.50000	—	0.50000	0.50000	—
β_S	Skilled bargaining	0.50000	0.50000	—	0.50000	0.50000	—
σ_S	OJS flow cost	1.10000	1.10000	—	1.10000	1.10000	—

Table 7: Parameter Estimates (B): Regime-Specific

		Financial Crisis			COVID		
Parameter	Description	Baseline	Crisis	Δ	Baseline	Crisis	Δ
Regime-specific							
P_U	Unskilled productivity	2.04377	1.90910	-0.13466	2.73686	2.09363	-0.64323
P_S	Skilled productivity	3.75855	4.05600	+0.29746	3.95147	4.95622	+1.00476
α_U	Unskilled damage shape	5.89456	5.94021	+0.04565	5.43224	7.78774	+2.35551
a_Γ	Skilled offer shape	2.75946	2.91353	+0.15407	3.39435	2.14112	-1.25323
b_Γ	Skilled offer shape	0.70915	1.14897	+0.43981	1.00198	0.82445	-0.17753
k_U	Unskilled vacancy cost	0.00979	0.00351	-0.00628	0.00207	0.00100	-0.00107
λ_U	Unskilled damage rate	0.29197	0.18142	-0.11054	0.32073	0.92715	+0.60642
k_S	Skilled vacancy cost	0.17944	0.19874	+0.01930	0.16986	0.92440	+0.75454
ξ_S	Skilled exog. sep. rate	0.00000	0.00000	+0.00000	0.00000	0.00000	-0.00000
λ_S	Skilled quality shock	0.42931	0.38727	-0.04204	0.43895	0.49556	+0.05661

2.3 Parameter Estimates

2.4 Crisis Assessment

The second-stage estimation fixes the deep structural parameters at their baseline values and re-estimates the ten regime-specific parameters on each crisis window. The identifying assumption is that preferences, matching technology, and bargaining institutions did not change between the expansion and the recession—only the aggregate state did. Tables 6 and 7 report the full set of estimates across all four windows. I discuss the regime-specific shifts for each crisis in turn, then compare the two episodes.

Productivity shifters. In both crises, unskilled productivity P_U falls while skilled productivity P_S rises. During the financial crisis the unskilled decline is moderate (-0.13, or -6.6%), while during COVID it is much larger (-0.64, or -23.5%). The rise in P_S is likewise more pronounced in COVID ($+1.00$, or $+25.4\%$) than in the financial crisis ($+0.30$, or $+7.9\%$). Because P_U and P_S are identified primarily by wage levels, the divergence reflects the asymmetric impact of each crisis on the two sectors. The unskilled market—dominated by service, retail, and construction jobs—bore the direct brunt of both the housing bust and the lockdowns. The skilled market proved more resilient, and the rise in P_S is consistent with the rapid adoption of remote work and digital technologies that raised the marginal product of educated workers during both downturns. The widening gap between P_S and P_U during COVID, in particular, matches the well-documented *K-shaped* recovery in which high-skill sectors rebounded quickly while low-skill sectors stagnated.

Unskilled damage shocks. The two crises generate sharply different patterns in the unskilled shock process. During the financial crisis, the damage arrival rate λ_U *falls* from 0.292 to 0.181 (-38%), while the damage shape α_U is essentially unchanged ($+0.05$). The decline in λ_U is counterintuitive if one expects recessions to increase the frequency of adverse match-specific draws. However, λ_U is jointly identified with the reservation threshold $\underline{p}_U(x, z)$ through the unskilled separation rate: what matters for separations is the product $\lambda_U \cdot G_z(\underline{p}_U)$. A decline in P_U raises the outside-option value relative to match output, pushing up the reservation threshold and increasing the probability of endogenous destruction conditional on any shock. The estimator compensates by lowering λ_U to prevent the separation rate from overshooting its target.

During COVID, the pattern reverses: λ_U nearly triples ($0.321 \to 0.927$, $+189\%$), but α_U rises sharply as well ($5.43 \to 7.79$, $+43\%$). Since post-shock match quality follows a Beta$(\alpha_U, 1)$ distribution, a higher α_U concentrates the post-shock draw closer to 1, making each individual

shock less damaging. COVID is thus characterised by *very frequent but individually milder* damage shocks in the unskilled market—consistent with the wave-like pattern of lockdowns and reopenings, in which unskilled matches were repeatedly disrupted but often at reduced intensity (e.g. capacity restrictions rather than full closures).

Skilled quality shocks. The skilled shock arrival rate λ_S moves modestly in both crises: a small decline during the financial crisis (-0.04) and a small increase during COVID ($+0.06$). Neither shift is large enough to indicate a fundamental change in the pace of skilled-match revaluation. This is consistent with the relative insulation of the skilled market from both downturns.

Vacancy costs. The unskilled vacancy cost k_U falls in both crises (FC: -0.006; COVID: -0.001). Although a rise in vacancy costs would be the textbook prediction for a demand-driven recession, the estimated decline can be rationalised through the free-entry condition. During a downturn, the vacancy-filling rate $q_U(\theta_U)$ rises as tightness falls and the pool of unemployed job seekers expands. If the expected match surplus does not fall proportionally—because, for instance, the reservation threshold adjusts upward and only the most productive matches survive—then free entry can be satisfied at a lower posting cost. The decline in k_U therefore absorbs the compositional improvement in the quality of new hires that accompanies a slack labour market.

The skilled vacancy cost k_S rises in both crises, but the magnitudes differ dramatically. During the financial crisis the increase is negligible ($+0.02$, or $+11\%$), whereas during COVID it is enormous ($0.170 \to 0.924$, a $4.4\times$ increase). This is the single largest parameter shift in either crisis and captures the unprecedented disruption to skilled hiring during the pandemic. Lockdowns, travel restrictions, and the abrupt transition to remote recruitment raised the effective cost of posting and filling skilled vacancies far beyond anything observed during the financial crisis.

Exogenous skilled separation. The exogenous skilled separation hazard ξ_S remains at zero in all four regimes. The model attributes all skilled-market job destruction to endogenous separation via the reservation rule $\underline{p}_S^*(x, z)$ and quality-shock redraws, never to exogenous layoffs unrelated to match-specific productivity. This is a strong finding: it implies that even during the pandemic—when large-scale layoffs were widespread—the estimated shock process is rich enough to rationalise the observed separation patterns without an additional exogenous margin.

Skilled offer distribution. The skilled offer distribution $\Gamma_z = \text{Beta}(a_\Gamma, b_\Gamma)$ shifts differently across the two crises. During the financial crisis, both parameters increase (a_Γ: +0.15; b_Γ: +0.44), with the rise in b_Γ dominating. Since a higher b_Γ pulls mass toward the lower end of the support, the implied mean offer quality falls from $a/(a+b) = 0.80$ to 0.72, and the distribution becomes more concentrated—the variance of available skilled-match quality narrows. The financial crisis thus degraded the average quality of skilled job offers while compressing their dispersion.

During COVID, both parameters fall (a_Γ: −1.25; b_Γ: −0.18), with the decline in a_Γ dominating. The mean offer quality falls from 0.77 to 0.72—a similar decline as in the FC—but the variance *increases* substantially (from 0.033 to 0.051). The COVID crisis therefore generated a more dispersed distribution of skilled-match quality: some firms offered very good matches (those that thrived during the pandemic), while others offered poor ones (those struggling to adapt), widening the spread of outcomes for trained workers entering the skilled market.

Comparing the two crises. Table 8 summarises the qualitative patterns across the two episodes.

Table 8: Qualitative Comparison of Regime-Specific Shifts

Channel	Financial Crisis	COVID
Unskilled productivity P_U	Moderate decline (−7%)	Large decline (−24%)
Skilled productivity P_S	Moderate rise (+8%)	Large rise (+25%)
Unskilled shock frequency λ_U	Falls (−38%)	Triples (+189%)
Unskilled shock severity α_U	Unchanged	Milder (higher α_U)
Skilled shock frequency λ_S	Small decline	Small rise
Unskilled vacancy cost k_U	Falls (−64%)	Falls (−52%)
Skilled vacancy cost k_S	Small rise (+11%)	Large rise (+444%)
Exogenous skilled separation ξ_S	Zero	Zero
Skilled offer quality (mean)	Falls	Falls
Skilled offer quality (dispersion)	Narrows	Widens

The financial crisis emerges as a relatively symmetric demand shock with moderate parameter shifts: both sector productivities move modestly, shock processes barely change, and vacancy costs adjust by small amounts. The COVID crisis, by contrast, is a far more asymmetric and disruptive event. The gap between skilled and unskilled productivity widens dramatically, unskilled matches are bombarded by frequent (if individually milder) shocks, and the cost of posting skilled vacancies increases more than fourfold. The two crises also differ in what they do to the skilled offer distribution: the financial crisis compressed it, while COVID dispersed it. These differences are economically plausible—the financial crisis was a broad

credit-driven contraction that affected most sectors similarly, while the pandemic created sharp winners and losers across industries and occupations, generating the heterogeneity reflected in the parameter estimates.

2.5 Training Cutoff

[PLACEHOLDER: Compare the training cutoff $\bar{x}(z_0)$ under the baseline parameters with $\bar{x}(z_1)$ under the crisis parameters. Report both values. A lower $\bar{x}$ during the crisis means that more workers find it optimal to train—the model generates countercyclical enrolment endogenously.

Connect to the empirical literature: Dellas and Sakellaris (2003), Barr and Turner (2015), and Blom et al. (2021) document that enrolment rises during recessions. The model's mechanism is a decline in the unskilled contact rate $\kappa_U(z)$, which compresses the continuation value of remaining unskilled and makes training relatively more attractive.]

2.6 Transition Dynamics

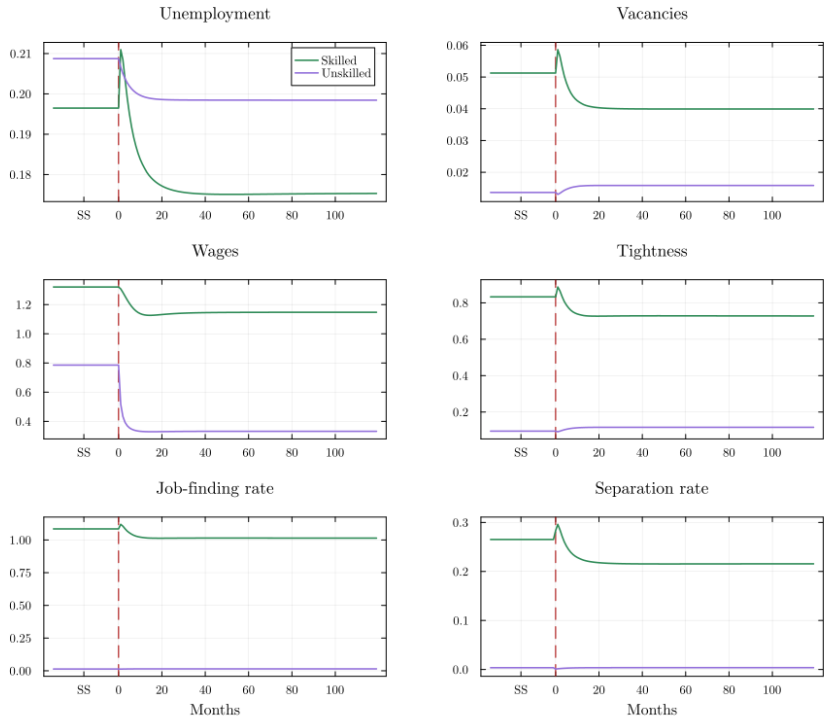

Figure 5: Nodel transition

3 Education-Subsidy Policies

The preceding sections used the estimated model to interpret two crises. This section turns the model toward policy. I take one of the estimated baselines, hold all parameters fixed, and ask: how does the stationary equilibrium respond to two alternative education subsidies?

3.1 Motivation

Search frictions create a hold-up problem for human-capital investment: because wages are set after training is complete, workers underinvest relative to the social optimum (Acemoglu and Shimer, 1999). The government can correct this distortion by subsidising the training margin, but the design of the subsidy matters. Two instruments target the same margin through different channels.

Policy A: higher training flow payoff b_T. The government provides monthly stipends to workers currently enrolled in training—living-cost grants or tuition waivers that raise disposable income during the training spell. In the model, this is a direct increase in the flow value b_T while the worker occupies the training state. It makes training more attractive by reducing the per-period cost of non-employment during the spell. The transfer is temporary: it stops once training is complete.

Policy B: lower training cost $c(x)$. The government invests in higher-education quality—better equipment, instructors, and facilities—at no additional cost to students. The subjective disutility of studying falls at every ability level. In the model, this is captured as a proportional reduction in the one-time cost function $c(x) = a_c\, e^{-x}$, replacing a_c with $(1-\delta)\, a_c$ for a subsidy rate $\delta \in (0,1)$. The cost reduction is permanent and lowers the up-front hurdle to initiating training.

3.2 Key Distinction

Both policies shift the training cutoff $\bar{x}(z)$ downward—more workers choose to train—but they operate through different channels. Policy A raises the value of the training state $T(x,z)$ by increasing b_T in equation (**??**). Because $T(x,z)$ feeds into the overall value of unskilled unemployment through equation (**??**), the net gain from training $-c(x) + T(x,z)$ rises at every ability level, and the indifference point shifts left. Policy B lowers $c(x)$ directly, making the training option $-c(x) + T(x,z)$ more attractive without changing the value of the training state itself.

The distinction matters for general-equilibrium feedbacks. Under both policies, the lower cutoff sends more workers through the training pipeline into skilled unemployment. The resulting increase in the effective pool of skilled seekers affects skilled-market tightness $\theta_S(z)$ through the free-entry condition (**??**). Whether tightness rises or falls depends on the vacancy-creation response: if firms post enough additional skilled vacancies to absorb the new entrants, the skilled contact rate $\kappa_S(z)$ may actually increase, reinforcing the subsidy. If not, congestion in the skilled market partially offsets the direct policy effect. This tightness feedback is the key general-equilibrium channel that competitive models of education policy—such as Heckman et al. (1998) and Abbott et al. (2019)—cannot capture.

3.3 Exercise Design

For each policy, I solve the new stationary equilibrium at three subsidy intensities—10%, 25%, and 50%—holding all other parameters at their estimated baseline values. For Policy A, each intensity corresponds to a percentage increase in b_T; for Policy B, to a percentage reduction in a_c.

3.4 Outcomes

For each policy and intensity level, I report the new stationary-equilibrium values of the training cutoff, the training stock, the skilled share, unemployment rates and market tightness in both sectors, mean wages, transition rates, and worker welfare at various ability levels.

[TODO: Insert results tables and figures here once the policy counterfactuals are computed.]

3.5 Discussion

Several questions guide the interpretation of the results.

Cost-effectiveness. Policy A carries a recurring fiscal cost—stipends paid every period for every trainee—that scales with both the training stock and the duration of the training spell. Policy B carries a one-time capital cost that permanently lowers $c(x)$. Even if both policies shift the cutoff by the same amount, their fiscal implications differ.

Diminishing returns. At high subsidy rates, one policy may saturate faster than the other. If the marginal worker brought into training by a large b_T increase has low ability

and therefore a long expected training duration, the per-worker cost of Policy A grows disproportionately. Policy B, by contrast, lowers the cost function uniformly, and its marginal effect on the cutoff may decline more slowly.

General-equilibrium feedbacks. The inflow of trained workers into the skilled market depresses skilled wages and may tighten the skilled market enough to partially offset the subsidy. Whether this dampening is quantitatively important depends on the elasticity of vacancy creation, which the free-entry condition pins down from the estimated parameters.

Baseline dependence. The ranking of the two policies may depend on the current state of the economy—whether the baseline is the pre-financial-crisis or the pre-COVID expansion. This interaction with the regime analysis is not a full regime-change exercise but a comparison of how the same subsidy performs under different initial conditions.

The connection to the empirical programme-evaluation literature is direct. Card et al. (2018) find that training programmes generate larger effects in the medium run than in the short run—a pattern consistent with the job-ladder dynamics in the skilled sector, where newly trained workers enter at the bottom of the match-quality distribution and climb over time. Lindner and Reizer (2020) show that the timing of benefit payments—flow versus frontloaded—generates different search responses holding total transfers constant, providing reduced-form motivation for distinguishing Policy A from Policy B.

4 Conclusion

This paper developed and estimated a structural search model in which workers endogenously choose whether to train and enter a higher-skill labour market. The model nests two segmented markets—an unskilled sector with endogenous job destruction and a skilled sector with on-the-job search and a job ladder—linked by a training cutoff that responds to wages, contact rates, and non-employment payoffs in both sectors.

Five results stand out.

First, training decisions respond endogenously to labour-market conditions, and the direction of the response depends on the nature of the shock. The training cutoff is not a fixed parameter: it shifts with aggregate conditions through the skilled wage premium, the probability of finding a skilled job after training, and the value of the unskilled outside option.

Second, a demand shock and a supply-and-reallocation shock operate through different chan-

nels and generate opposite predictions for training take-up. The financial crisis depressed both sectors roughly symmetrically, leaving the net gain from training largely unchanged. COVID disproportionately damaged the unskilled sector while expanded unemployment benefits cushioned the training spell, lowering the cutoff and drawing more workers into training.

Third, the multi-regime estimation strategy—estimating pre-crisis, fixing deep parameters, re-estimating only regime-specific parameters during crisis—identifies precisely which features of the aggregate state shifted. The structural stability check across the two baselines confirms that the deep parameters are internally consistent across expansions.

Fourth, transition dynamics after each shock reveal that the adjustment of the skilled share is slow, driven by the pipeline delay between training initiation and labour-market entry. Cyclical shocks can therefore generate long-run hysteresis in skill composition even when the shock itself is temporary.

Fifth, education subsidies that target the same margin—the training cutoff—through different channels produce different outcomes once general-equilibrium feedbacks are accounted for. Raising the training flow payoff (stipends) and lowering the one-time training cost (institutional investment) both shift the cutoff, but they differ in fiscal cost, in how they interact with the training pipeline, and in the tightness response they induce in the skilled market.

The model abstracts from several features that could enrich the analysis. Within-market human capital accumulation—allowing a skill multiplier to evolve with experience—would add a second margin of adjustment that interacts with the training decision. Heterogeneous firms would allow the model to speak to the sorting patterns documented by Lise et al. (2016) and Lise and Robin (2017). Temporary regime changes, rather than permanent switches, would better match the observed recovery dynamics after each crisis. These extensions are left for future work.

The broader message is that structural models which nest regime changes and policy interventions provide a framework for crisis-contingent and investment-oriented labour-market policy. The nature of the crisis determines how workers reallocate across markets, and the design of the subsidy determines whether the reallocation is reinforced or dampened by general-equilibrium feedbacks.

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